

Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Section 1: Case Summary

Scenario Title:	Spinal Cord Injury with Respiratory Failure
Keywords:	Respiratory Failure, Spinal Cord Injury, Flail Chest
Brief Description of Case:	A 46- year-old female pedestrian struck by a truck is admitted to ICU with multiple injuries, including a thoracic spinal cord injury, a right flail chest and pneumothorax, and a liver laceration. While awaiting OR for spinal stabilization, she decompensates into mixed respiratory failure and hemodynamic instability. She will require ventilatory support, escalation of vasopressors, and will develop a tension pneumothorax post-intubation.

Goals and Objectives	
Educational Goal:	Manage respiratory failure in a patient with Spinal Cord Injury.
Objectives: (Medical and CRM)	1. Manage mixed respiratory failure in a trauma patient with multiple injuries. 2. Identify and manage neurogenic shock in a trauma patient. 3. Consider a differential for hypoxia and shock post-intubation. 4. Optimize resource utilization during medical crises. 5. Resuscitate a patient in a challenging, unfamiliar environment. 6. Seek help from consultants when faced with a deteriorating patient requiring definitive management.
EPAs Assessed:	Core EPA: C1 Resuscitating and coordinating care for critically ill patients.

Learners, Setting and Personnel			
Target Learners:	☐ Junior Learners	☒ Senior Learners	☐ Staff
	☒ Physicians	☒ Nurses	☒ RTs
	☐ Other Learners:		
Location:	☒ Sim Lab	☒ In Situ	☐ Other:
Recommended Number of Facilitators:	Instructors: 1		
	Sim Actors: 2		
	Sim Techs: 1		

Scenario Development	
Date of Development:	May 21, 2022
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Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Aisha Tahan			Age: 46	Gender: female	Weight: 65 kg
Reason for Referral: Major blunt trauma					
Temp: 37.1	HR: 48	BP: 79/48	RR: 30	O ₂ Sat: 93%	FiO ₂ : 100% on NRB
Cap glucose: 8			GCS: 9 (E2 V2 M5)		
Referral Information: 46-year-old female pinned between truck and car resuscitated in ED with multiple injuries, including bilateral rib fractures with flail, liver laceration, and unstable T5-6 fractures with reduction of canal diameter. ICU has admitted the patient while she awaits OR later today to stabilize the spine.					
Allergies: none					
Past Medical History: none			Current Medications: Levophed Hydromorphone Tylenol Gabapentin		

Section 2B: Extra Patient Information

A. Further History
<i>The following information should be provided by the sim actor playing the ICU Resident, who will call the ICU Fellow to begin the case. They will relay an abridged version of this information via phone, and then the ICU Fellow (learner) enters the room to begin the sim.</i> Mrs. Tahan is a 46-year old woman who was hit by a truck backing up and who got pinned between the truck and another car. She was conscious at the scene but reported loss of sensation and movements of her legs. She was brought to the ER by EMS. A litre of ringer's lactate was administered by EMS. During the primary survey, she had a low BP and received one unit of RBCs with a response of her SBP to ~90. A chest tube was inserted on the right side. The ICU first assessed the patient after completion of trauma imaging. The following injuries have been identified: - No brain injury; no cervical spine injury - Bilateral rib fractures right #7-11 with flail segment; left #8-10; lower sternal fracture - Small right residual pneumothorax after chest tube placement, occult left pneumothorax - Grade 2 liver laceration - Thoracic spine T#5-6 displaced, unstable fractures with 50% reduction of canal diameter - L1-L2 traverse process fractures - No pelvic and no long bone fracture At the time of initial ICU assessment earlier today, the patient was alert, oriented, and reported moderate thoracic pain (6/10). Her VS were: BP 88/50, HR 51/min, Sat 93% on Venturi mask 50%, RR 25/min, Temp 36.9. The ICU ordered an infusion of levophed for a MAP >80, analgesia (IV hydromorphone + Tylenol + Gabapentin), and an admission to a level 2 ICU. The Spine Team was organizing OR time later today to stabilize the spine.



Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

The remaining information below should be relayed by the ICU resident in full detail at the bedside.

The ICU resident was just called to reassess the patient due to increasing work of breathing, fluctuating LOC, and increasing vasopressor requirements. VS are now:

BP 79/48 on norepinephrine at 15ug/min, HR 48/min, Sat 93% on NRM, RR 30/min, Temp 37.1, GCS 9 (E2, V2, M5)

The ICU resident sent an ABG, gave 500cc of NS, ordered a chest x-ray, and called the ICU fellow for immediate assistance.

B. Physical Exam

List any pertinent positive and negative findings

Cardio: Tachycardic, hypotensive on Levo. Extremities warm.

Neuro: GCS 9 (E2 V2 M5)

Resp: Tachypneic and hypoxic. Moderate work of breathing, with right-sided flail chest. Equal air entry bilaterally.

Head & Neck: no signs of trauma

Abdo: Bruised lower thorax and upper right abdomen. Soft, nondistended, nontender

MSK/skin: no pelvic or long bone fractures

Other:

Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (adult)
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Arterial line Central Line (Right IJ Cordis) ECG leads NIBP Peripheral IVs (2 in place) Pulse Oximeter Intubation Supplies / Airway Cart Ventilator Chest tube (manikin right chest) Chest tube tray (available) Crash cart
C. Required Medications
Vasopressors (running NE) ACLS medications
D. Moulage
Bruised right lower thorax and upper abdomen
E. Monitors at Case Onset
☒ Patient on monitor with vitals displayed ☐ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A – C-Collar in place. Patient moaning when stimulated. No obvious trauma to face or neck. B – Moderate work of breathing. Equal air entry bilaterally, no crackles or wheeze. Right-sided flail chest. C – HR 48 BP 79/48 on Levo. Warm extremities. Bruising to abdomen, but soft, nontender, nondistended. Pelvis stable, no long-bone deformities. D - GCS 9 (E2 V2 M5). E – T37.1

Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Section 4: Sim Actor and Standardized Patients

Sim Actor and Standardized Patient Roles and Scripts	
Standby Roles	(available if called) 1. RT 2. Spine fellow (by phone) 3. Trauma fellow (by phone) 4. 2nd ER RN 5. 2nd RT
ICU Resident Script	You have been asked to go reassess this patient in the ER, and found the patient in respiratory failure, on moderate dose of vasopressors, with a fluctuating LOC. Currently: BP 79/48 on norepinephrine at 15ug/min, HR 48/min, Sat 93% on NRM, RR 30/min, Temp 37.1 GCS 9 (E2, V2, M5) You sent an ABG, gave 500cc of NS, ordered a chest x-ray, and just called the fellow for immediate assistance. Please assist the fellow and perform at the level of a PGY2 GIM resident.
ER RN Script	The Spine Team is organizing OR time later today to stabilize the spine as a B case, but you haven't heard about an OR time. You just called back the ICU resident to reassess this patient in the ER, as she looks more short of breath, requiring higher moderate dose of vasopressors, and has a lower, fluctuating LOC. The resident just called the fellow for immediate assistance. Currently: BP 79/48 on norepi at 15ug/min, HR 48/min, Sat 93% on NRM, RR 30/min, Temp 37.1 GCS 9 (E2, V2, M5) Please assist the ICU team with the resuscitation.

Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus Bradycardia HR: 48 BP: 79/48 RR: 30 O ₂ SAT: 93% on 100% NRB T: 37.1°C GCS: 9 E2V2M5	<i>Patient is confused, and appears to be in respiratory distress.</i>	<u>Expected Learner Actions</u> ☐ Consider multiple causes of shock in a trauma patient ☐ Perform FAST scan to assess for intraabdominal free fluid ☐ Assess for hemo/pneumothorax using POCUS or CXR ☐ Increase norepinephrine ☐ Add 2 nd vasopressor ☐ Add epinephrine for bradycardia	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Increased norepinephrine => Increased MAP - 2 nd vasopressor added => MAP above 65 - IV bolus, blood transfusion => no response - Add epinephrine => HR 65 <u>Triggers</u> <i>For progression to next state</i> - MAP > 65	- labs and imaging may be provided if the learner asks for them - POCUS shows no pericardial effusion, good lung sliding bilaterally, and no intraabdominal free fluid (<i>QR images should be attached to manikin</i>).
2. Respiratory Failure Rhythm: Sinus HR: 65 BP: 95/65 RR: 30 O ₂ SAT: 93% on 100% NRB T: 37.1°C GCS: 9 E2V2M5	<i>Despite improved hemodynamics, the patient becomes more confused and continues to have respiratory distress.</i>	<u>Expected Learner Actions</u> ☐ Consider a trial of NIPPV ☐ Recognize need for ETI ☐ Vocalize an airway plan with contingencies ☐ Maintain spinal precautions while intubating ☐ Intubate the patient ☐ Call consultants for help given patient's deteriorating clinical status	<u>Modifiers</u> - Oral Airway => increased cough - BVM => no improvement in sats unless PEEP valve at 10 - Optiflow / BiPAP => patient becomes agitated and removes device <u>Triggers</u> - Endotracheal intubation	



Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

3. Post-Intubation Deterioration Rhythm: Sinus HR: 65 BP: 60/40 RR: dictated by vent O2SAT: 80% intubated on 100% FiO2 T: 37.1°C GCS: 3 (sedated and intubated)	<i>2 minutes after intubation, the patient's oxygen saturation and blood pressure drop</i>	<u>Expected Learner Actions</u> ☐ Consider a focused differential for hypotension and hypoxia post-intubation ☐ Disconnect the ET tube from the vent and manually ventilate the patient ☐ Make the diagnosis of left tension pneumothorax with clinical exam or (preferably) POCUS ☐ Treat left tension pneumothorax with either needle decompression or chest tube insertion ☐ If not yet done, call consultants for help	<u>Modifiers</u> - Disconnecting the ET tube from the vent => no change, easy manual ventilation with unequal chest rise - Delay to management => progressive hypoxia and ultimate arrest <u>Triggers</u> - Chest tube or finger thoracostomy => normalization of O2 sat and BP	- Manikin should have decreased air entry to the left lung. - POCUS shows absent lung sliding on the left side. If repeated, cardiac POCUS shows no pericardial effusion, and stable trace intraabdominal free fluid. - No intervention except decompressing the chest will improve clinical status. - If the learner calls for a stat CXR and elects to wait for it, the patient should become progressively hypoxic and go into a PEA cardiac arrest. If this happens, the learner should rapidly decompress the chest, which will result in ROSC within 1 round of CPR.
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Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Appendix A: Laboratory Results

<u>CBC</u> WBC 16.8 Hgb 99 Plt 210 <u>Lytes</u> Na 142 K 3.8 Cl 105 HCO ₃ 19 AG 17 Urea 6.5 Cr 95 Glucose 9.9 CK 345 <u>Extended Lytes</u> Ca 2.06 Mg 0.74 PO ₄ 1.81 Albumin 37 <u>ABG</u> pH 7.28 pCO ₂ 44 pO ₂ 100 HCO ₃ 19 Lactate 2.6	<u>Cardiac/Coags</u> Trop 32 INR 1.2 aPTT 30 <u>Biliary</u> AST 40 ALT 79 GGT 50 ALP 115 Bili 8 Lipase 100 <u>Tox</u> EtOH 0 <u>Other</u> B-HCG negative
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Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Appendix B: ECGs, X-rays, Ultrasounds and Pictures

X-RAY TRAUMA SERIES ER

Chest: Patient is mildly rotated. Bilateral rib fractures with sternal fracture. Right chest tube in place. Small apical residual right pneumothorax. No left pneumothorax.

Pelvis: No fracture

CT SCAN HEAD & NECK

PRELIMINARY REPORT

No traumatic brain injury. No acute fracture of cervical spine.

CT SCAN CHEST

PRELIMINARY REPORT

Bilateral rib fractures, including right #7-8-9 with flail component, and left #8-10 (no flail). Lower sternal fracture. Right chest tube in place. Small residual right pleural pneumothorax. Occult left pneumothorax. No aortic injury. No pericardial effusion.

CT SCAN ABDOPELVIS

PRELIMINARY REPORT

Grade 2 liver laceration. No significant bowel wall thickening. No pneumatosis. No free gas in the abdomen. There is inter-loop mesenteric free fluid in the right lower quadrant.

CT SCAN SPINE

PRELIMINARY REPORT

Thoracic spine T#5-6 displaced, unstable fractures with 50% reduction of canal diameter. L1-L2 transverse process fractures

Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

POCUS IMAGES:

Normal lung sliding:



Normal heart (no PCE):



Normal RUQ:



Pneumothorax:



Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Medical management questions (see cheat sheet below)

1. What was the cause of this patient's hypotension? How does neurogenic shock differ from hemorrhagic shock, in presentation and management?
2. What specific considerations are there for managing the airway in a patient with SCI?
3. What was the cause of this patient's respiratory failure?
4. What is your differential diagnosis for post-intubation hypotension? Hypoxia?

Crisis-resource management questions:

5. What considerations did you have in calling consultants for this patient?

Cheat Sheet:

1. What was the cause of this patient's hypotension? How does neurogenic shock differ from hemorrhagic shock, in presentation and management?
 - a. This patient has multiple reasons for possible hypotension, including hemorrhagic shock given her liver laceration and neurogenic shock from her spinal cord injury (which is the main reason for her hypotension).
 - b. Neurogenic shock is a type of vasodilatory shock. The key clinical differentiators between neurogenic and hemorrhagic shock are the warm vs cool extremities, bradycardia vs tachycardia, and normal urine output vs reduced urine output, respectively.
 - c. Neurogenic shock may be seen in SCI above T6. The cause is the loss of sympathetic tone to the peripheral vasculature and heart, causing vasodilation and bradycardia (although bradycardia may not be present, and is not required for the diagnosis). It should be a diagnosis of exclusion, once other causes of shock in trauma have been ruled out (hemorrhagic, obstructive, etc).
 - d. A higher MAP should be targeted in neurogenic shock than hemorrhagic, typically MAP > 85-90 to preserve perfusion of the cord and prevent secondary injury. This should be achieved with IV fluids and vasopressors, if required. Norepinephrine is an appropriate first line agent. Atropine may also be used for bradycardia.
2. What specific considerations are there for managing the airway in a patient with SCI?
 - a. Many SCI patients require intubation, especially those with cervical injury; complete cord injuries above C5 should be intubated prophylactically, given the diaphragmatic weakness. High C-spine injuries above C3 result in rapid respiratory and subsequent cardiac arrest unless quickly intubated. Traumatic arrest in the field without an identified cause should prompt consideration of a high C-spine injury.
 - b. Intubation may also be indicated in the setting of lower levels of SCI, due to thoracic muscle weakness (discussed below). Increasing work of breathing should prompt consideration for intubation. Need to transport the patient to a second facility may also be an indication for prophylactic intubation.
 - c. In the setting of trauma, there may be associated injuries to the face or neck that may additionally compromise the airway. The airway will be inherently difficult due to the necessity for spinal immobilization. One individual must maintain the C-spine in a neutral position while removing the collar and during intubation, which should be performed by a second individual.



Critical Care Case Series 3: Spinal Cord Injury with Respiratory Failure

3. What was the cause of this patient's respiratory failure?
 - a. This is likely multifactorial, including her flail chest, pneumothorax, and spinal cord injury.
 - b. SCI can cause respiratory failure due to muscle weakness i.e. paradoxical abdominal breathing. The diaphragm is innervated by C3-C5, and the intercostal muscles are innervated by T1-11. Even a lower SCI, as in this patient, can therefore cause respiratory failure due to loss of the intercostals leading to reduced vital capacity, atelectasis, and hypoxia. Paralysis of the intercostals leads to chest wall contraction with activation of the diaphragm, leading to paradoxical abdominal breathing and a significant drop in ability to ventilate.
 - c. Additionally, the external intercostals and abdominal wall muscles assist with coughing. Abdominal wall muscles originate from T6-12. Paralysis of these muscles can lead to weak cough and inability to clear secretions, leading to aspiration. This may be exacerbated in acute SCI because unopposed vagal activity may cause increased secretions and bronchial constriction.
4. What is your differential diagnosis for post-intubation hypotension? Hypoxia?
 - a. Post-intubation hypoxia has a short differential that should be approached in a structured fashion. A mnemonic to remember it is **DOPES**: Displacement of the ET tube, Obstruction of the tube, Patient (pneumothorax, PE), Equipment failure, and Stacked breaths. A structured approach:
 - i. Disconnect the ET tube from the ventilator and manually ventilate the patient. This allows for release of trapped breaths, and will identify if the problem is with the ventilator.
 - ii. Assess for equal chest rise, oxygen saturation, cyanosis, and hemodynamic stability.
 - iii. If the patient is difficult to manually ventilate, quickly identify if the issue is with the tube. Check the EtCO₂, pass a suction catheter through the ET tube, and visually confirm the placement of the tube with laryngoscopy or bronchoscopy. If in doubt, replace the tube.
 - iv. If there is good chest rise, the problem is unlikely the ET tube. Look for asymmetric chest rise (consider mainstem intubation and pneumothorax), assess for pulmonary edema or bronchospasm, and consider PE.
 - v. If there is little resistance to manual ventilation, but there is minimal chest movement, consider a circuit leak, or dislodged tube (i.e. esophageal intubation).
 - vi. If the patient is easy to manually ventilate and the hypoxia rapidly resolves, consider a problem with the ventilator.
 - b. Post-intubation hypotension has a broader differential. Consider underlying patient factors, like acidosis and hypovolemia. Specific cardiorespiratory problems likely to be worsened by positive-pressure ventilation include cardiac tamponade, pulmonary hypertension / pulmonary embolism, pneumothorax, and breath stacking / bronchospasm. Anaphylaxis to any drugs administered should be considered, as well as hyperkalemia if succinylcholine is used. Finally, induction agents can cause hypotension.

References

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